

MAT 324 HW04

Aditya Pahuja

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Problem 1. Prove or give a counterexample: If $F \subseteq \mathbb{R}$ is such that every bounded continuous function from F to $\mathbb{R}$ can be extended to a continuous function from $\mathbb{R}$ to $\mathbb{R}$, then F is a closed subset of $\mathbb{R}$.

Solution. Suppose $F \subseteq \mathbb{R}$ is not closed; then, F does not contain one of its limit points $p \in \mathbb{R}$. This means we can find a strictly monotone sequence $x_1, x_2, \dots \in F$ converging to p ; suppose without loss of generality that this sequence is increasing. Define $g: F \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} (-1)^n & x = x_n \text{ for some integer } n \\ tg(x_n) + (1-t)g(x_{n-1}) & x = tx_n + (1-t)x_{n-1} \text{ for some } t \in [0, 1] \text{ and some } n \in \mathbb{N} \\ 1 & \text{otherwise.} \end{cases}$$

This function is continuous on $(-\infty, p) \cap F$ by the pasting lemma because it is continuous on the locally finite collection of closed intervals $(-\infty, x_1]$ and $[x_n, x_{n+1}]$ for $n \in \mathbb{N}$; the collection is locally finite because

- (1) it is of course locally finite on the interior of each of these intervals;
- (2) for any x_n , the neighborhood (x_{n-1}, x_{n+1}) intersects only $[x_{n-1}, x_n]$ and $[x_n, x_{n+1}]$ and no others in the collection.

Also, g is continuous on $(p, \infty) \cap F$ because it is constantly 1 on $(p, \infty) \cap F$. Therefore, g is continuous on F , and it is bounded between -1 and 1 .

However, g cannot be extended to a continuous function $\mathbb{R} \rightarrow \mathbb{R}$ because for any $\delta > 0$, the neighborhood $(p - \delta, p + \delta)$ contains the subsequence $x_N, x_{N+1}, \dots$ of $(x_n)_{n \in \mathbb{N}}$, which implies that g achieves both the values 1 and -1 in any open neighborhood of p ; that is, $\lim_{x \rightarrow p} g(x)$ doesn't exist, which means that any extension $\tilde{g}$ of g cannot be continuous at p .

That is, no non-closed subset F of $\mathbb{R}$ has the property that bounded continuous functions on F can be extended to continuous functions on $\mathbb{R}$, so contrapositively any set with the extension property must be closed in $\mathbb{R}$. $\square$

Problem 2. (a) Let $A, B_1, B_2, \dots \subseteq \mathbb{R}$ and $B = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} B_k$. Show that

$$B \setminus A \subseteq \bigcup_{k=n}^{\infty} (B_k \setminus A) \quad \text{and} \quad A \setminus B \subseteq \bigcup_{k=n}^{\infty} (A \setminus B_k)$$

for all $n \in \mathbb{N}$.

(b) Show that $A \subseteq \mathbb{R}$ is Lebesgue if and only if for every $\varepsilon > 0$ there exists a Borel subset $B \subseteq \mathbb{R}$ such that

$$|(B \setminus A) \cup (A \setminus B)| < \varepsilon.$$

Solution. (a) We can write

$$B \setminus A = \bigcup_{n=1}^{\infty} \left(\left(\bigcap_{k=n}^{\infty} B_k \right) \setminus A \right) = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} (B_k \setminus A)$$

Since $\bigcup_{k=n}^{\infty} (B_k \setminus A)$ is an increasing sequence of sets in n , we can get rid of the first N terms without changing the union; that is,

$$B \setminus A = \bigcup_{n=N}^{\infty} \bigcap_{k=n}^{\infty} (B_k \setminus A) \subseteq \bigcup_{n=N}^{\infty} \bigcup_{k=n}^{\infty} (B_k \setminus A) = \bigcup_{k=N}^{\infty} (B_k \setminus A)$$

for any $N \in \mathbb{N}$. For the other statement we similarly have

$$A \setminus B = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (A \setminus B_k) = \bigcap_{n=N}^{\infty} \bigcup_{k=n}^{\infty} (A \setminus B_k) \subseteq \bigcup_{n=N}^{\infty} \bigcup_{k=n}^{\infty} (A \setminus B_k) = \bigcup_{k=N}^{\infty} (A \setminus B_k).$$

for any $N \in \mathbb{N}$ because $\bigcup_{k=n}^{\infty} (A \setminus B_k)$ is a decreasing sequence in N so we can discard the first N terms in the intersection.

(b) If $A \subseteq \mathbb{R}$ is Lebesgue, then Axler 2.71c shows that there exists a Borel set $B \subseteq A$ (manifested as the union of countably many open subsets of $\mathbb{R}$) such that $|A \setminus B| = 0$. Of course $B \subseteq A$ implies $|B \setminus A| = 0$, so

$$|(B \setminus A) \cup (A \setminus B)| = 0 < \varepsilon \text{ for all } \varepsilon > 0.$$

Conversely, suppose that A is a subset of $\mathbb{R}$ such that given $\delta > 0$, there exists a Borel set B_δ such that $|(A \setminus B_\delta) \cup (B_\delta \setminus A)| < \delta$, and for brevity, let $D_\delta = (A \setminus B_\delta) \cup (B_\delta \setminus A)$. Then, let $\varepsilon > 0$. By definition of outer measure, there exists an open set $U \subseteq \mathbb{R}$ such that $D_{\varepsilon/3} \subseteq U$ and

$$|U| < \varepsilon/3 + |D_{\varepsilon/3}| < 2\varepsilon/3.$$

This means $B_{\varepsilon/3} \setminus U$ is a Borel set contained in A such that

$$\begin{aligned} |A \setminus (B_{\varepsilon/3} \setminus U)| &= |A \cap (B_{\varepsilon/3} \cap U^c)| = |A \cap (B_{\varepsilon/3}^c \cup U)| \\ &= |(A \setminus B_{\varepsilon/3}) \cup (A \cap U)| \leq |D_\delta| + |U| < \varepsilon/3 + 2\varepsilon/3 = \varepsilon. \end{aligned}$$

We showed in class that A is Lebesgue if and only if for each $\varepsilon > 0$ there exists a Lebesgue set $A_\varepsilon^- \subseteq \mathbb{R}$ such that $|A \setminus A_\varepsilon^-| < \varepsilon$. Here existence is shown by taking $A_\varepsilon^- = B_{\varepsilon/3} \setminus U$, which is sufficient to conclude that A is Lebesgue. $\square$

Problem 3. (a) Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $f: X \rightarrow [0, \infty]$ is an $\mathcal{S}$ -measurable function. Prove that

$$\int_X f \, d\mu > 0 \text{ if and only if } \mu(\{x \in X : f(x) > 0\}) > 0.$$

(b) Give an example of a Borel measurable function $f: [0, 1] \rightarrow (0, \infty)$ such that the lower Riemann integral $L(f, [0, 1])$ is 0.

(c) Compute $\int f \, d\lambda$, where f is your answer for (b) and λ is the Lebesgue measure on $[0, 1]$.

Solution. (a) Let $E = \{x \in X : f(x) > 0\}$ and $E_n = \{x \in X : f(x) > 1/n\}$. We have $\frac{1}{n}\mathbf{1}_{E_n} \leq f$, so

$$\frac{\mu(E_n)}{n} = \int_X \frac{1}{n}\mathbf{1}_{E_n} \, d\mu \leq \int_X f \, d\mu$$

Since $E = \bigcup_{n=1}^{\infty} E_n$, $\mu(E) = \lim_{n \rightarrow \infty} \mu(E_n)$, so if $\mu(E) > 0$ there exists N such that $\mu(E_N) > 0$. This implies that integral of f is positive because it is bounded below by $\frac{\mu(E_N)}{N}$.

Conversely suppose $\mu(E) = 0$. If P is a $\mathcal{S}$ -partition of X , then for $I \in P$, or equivalently $\inf_I f \neq 0$ if and only if $I \subseteq E$. Therefore

$$\mathcal{L}(f, P) = \sum_{I \in P} (\inf_I f) \mu(I) = \sum_{\substack{I \in P \\ I \subseteq E}} (\inf_I f) \underbrace{\mu(I)}_{\leq \mu(E)=0} = 0$$

for every $\mathcal{S}$ -partition of X , so taking the supremum gives $\int_X f \, d\mu = 0$.

(b) Define $f: [0, 1] \rightarrow (0, \infty)$ by

$$f(x) = \begin{cases} 1 & \text{if } x \notin \mathbb{Q} \\ 1/q & \text{if } x = p/q \in \mathbb{Q} \text{ with } p, q \in \mathbb{Z}_{\geq 0} \text{ and } \gcd(p, q) = 1 \end{cases}$$

where $\mathbb{Z}_{\geq 0}$ is the set of nonnegative integers (note that this means $f(0) = 1$). Then, any interval $I \subseteq [0, 1]$ with more than one point contains infinitely many rationals. Since there are only finitely many rational numbers with a given denominator, I contains rational numbers with arbitrarily large denominator, so

$$\inf_{x \in I} f(x) \leq \frac{1}{N}$$

for arbitrarily large N , i.e. $\inf_{x \in I} f(x) = 0$. This means that any interval I in a partition P of $[0, 1]$ contributes $0 \cdot \ell(I) = 0$ to the lower Riemann sum $L(f, P)$, so the supremum of $L(f, P)$ over all partitions P is $L(f, [0, 1]) = 0$.

It thus remains to verify that f is Borel measurable. Since the image of f is the set $\{1/n : n \in \mathbb{N}\}$, for any $S \subseteq (0, \infty)$, the set $f^{-1}(S)$ is the union of sets of the form $f^{-1}(1/n)$ for some $n \in \mathbb{N}$, each of which is Borel since $f^{-1}(1)$ is $[0, 1] \setminus \mathbb{Q}$, which is the complement of the Borel set $[0, 1] \cap \mathbb{Q} = \bigcup_{q \in [0, 1] \cap \mathbb{Q}} \{q\}$, and $f^{-1}(1/n)$ for $n > 1$ is finite. There are only countably many sets of the form $f^{-1}(1/n)$, so in fact $f^{-1}(S)$ for any $S \subseteq (0, \infty)$ is a countable union of Borel sets, hence is a Borel set.

(c) The integral of f with respect to λ is $\boxed{1}$ because

$$\begin{aligned} 1 &= \lambda([0, 1] \setminus \mathbb{Q}) = \int_{[0, 1]} \mathbf{1}_{[0, 1] \setminus \mathbb{Q}} \, d\lambda \\ &\leq \int_{[0, 1]} f \, d\lambda \leq \int_{[0, 1]} \mathbf{1}_{[0, 1]} \, d\lambda = \lambda([0, 1]) = 1, \end{aligned}$$

where the inequalities are due to $0 < f(x) \leq 1$ for rational x and $f(x) = 1$ for irrational x . $\square$

Problem 4. Suppose μ is the measure on $(\mathbb{N}, 2^{\mathbb{N}})$ defined by

$$\mu(E) = \sum_{n \in E} \frac{1}{2^n}.$$

Prove that for every $\varepsilon > 0$, there exists a set $E \subseteq \mathbb{N}$ with $\mu(\mathbb{N} \setminus E) < \varepsilon$ such that any sequence of functions $f_1, f_2, \dots : \mathbb{N} \rightarrow \mathbb{R}$ that converges pointwise on $\mathbb{N}$ also converges uniformly on E .

Solution. Given $\varepsilon > 0$, pick N such that $\frac{1}{2^N} < \varepsilon$ and set $E = \{1, \dots, N\}$, so that $\mathbb{N} \setminus E = \{N+1, N+2, \dots\}$ has measure

$$\sum_{n=N+1}^{\infty} \frac{1}{2^n} = \frac{1}{2^N} < \varepsilon.$$

Let $\varepsilon' > 0$. For each $k = 1, \dots, N$, pointwise convergence of the sequence $f_1(k), f_2(k), \dots$ means that there exists $M_k > 0$ such that $n > M_k$ implies $|f(k) - f_n(k)| < \varepsilon'$. Take $M = \max_{1 \leq k \leq N} M_k$; then $n > M$ implies $|f(k) - f_n(k)| < \varepsilon'$ for all $k \in E$. This is what it means for $f_1, f_2, \dots$ to converge uniformly to f on E , so we are done. $\square$

Problem 5. Let $(X, \mathcal{S}, \mu)$ be a measure space.

(a) Let $f_n : X \rightarrow [0, \infty]$ be a sequence of measurable functions. Show that

$$\int_X \left(\sum_{n=1}^{\infty} f_n \right) d\mu = \sum_{n=1}^{\infty} \left(\int_X f_n d\mu \right).$$

(b) Let $f : X \rightarrow [0, \infty)$ be a measurable function and

$$E_n = \{x \in X : 2^n < f(x) \leq 2^{n+1}\}$$

for all $n \in \mathbb{Z}$. Show that $\int_X f d\mu < \infty$ if and only if $\sum_{n \in \mathbb{Z}} 2^n \mu(E_n) < \infty$.

(c) Let $\alpha \in \mathbb{R}$ and λ be the Lebesgue measure on $(0, 1)$. Show that $\int_{(0,1)} x^{-\alpha} d\lambda < \infty$ if and only if $\alpha < 1$.

Solution. (a) Let $g_N = \sum_{n=1}^N f_n$ and $g = \lim_{N \rightarrow \infty} g_N = \sum_{n=1}^{\infty} f_n$. Since the f_n are all nonnegative we can see that the g_N are a sequence of increasing functions, so the monotone convergence theorem yields

$$\begin{aligned} \int_X \left(\sum_{n=1}^{\infty} f_n \right) d\mu &= \int_X g d\mu = \lim_{N \rightarrow \infty} \int_X g_N d\mu \\ &= \lim_{N \rightarrow \infty} \int_X \left(\sum_{n=1}^N f_n \right) d\mu = \lim_{N \rightarrow \infty} \sum_{n=1}^N \left(\int_X f_n d\mu \right) = \sum_{n=1}^{\infty} \left(\int_X f_n d\mu \right). \end{aligned}$$

(b) Observe that the E_n are pairwise disjoint and cover X , and let $g(x) = \sum_{n \in \mathbb{Z}} 2^n \cdot \mathbf{1}_{E_n}(x)$ be a simple function. Since $2^n < f(x) \leq 2^{n+1}$ for all $x \in E_n$, we have $g(x) \leq f(x) < 2g(x)$ for all $x \in X$. Therefore, if $\int_X f d\mu < \infty$, then

$$\sum_{n \in \mathbb{Z}} 2^n \mu(E_n) = \int_X g d\mu \leq \int_X f d\mu < \infty,$$

and if conversely the sum is finite, then

$$\int_X f d\mu \leq \int_X 2g d\mu = \sum_{n \in \mathbb{Z}} 2^{n+1} \mu(E_n) = 2 \cdot \sum_{n \in \mathbb{Z}} 2^n \mu(E_n) < \infty.$$

(c) Given $\alpha \in \mathbb{R}$ define

$$E_n = \{x \in (0, 1) : 2^n < x^{-\alpha} \leq 2^{n+1}\} = \{x \in (0, 1) : 2^{-(n+1)/\alpha} < x \leq 2^{-n/\alpha}\}.$$

If $\alpha = 0$ then $x^{-\alpha} = 1$ for all $x \in (0, 1)$ so the integral is $\int_{(0,1)} 1 \, d\lambda = 1 < \infty$.

If $\alpha < 0$, then $-\alpha > 0$ so $0 < x^{-\alpha} < 1$. This means that

$$\int_{(0,1)} x^{-\alpha} \, d\lambda < \int_{(0,1)} 1 \, d\lambda = \lambda((0,1)) = 1 < \infty.$$

If $\alpha > 0$, then since $x \in (0, 1)$, $2^{-(n+1)/\alpha} < x$ can only happen if n is nonnegative; otherwise, if $n \leq -1$, then $-(n+1)/\alpha \geq 0$ so $2^{-(n+1)/\alpha} \geq 1$. For these nonnegative n , $\lambda(E_n) = 2^{-n/\alpha} - 2^{-(n+1)/\alpha}$. The series

$$\begin{aligned} \sum_{n \in \mathbb{Z}} 2^n \lambda(E_n) &= \sum_{n=1}^{\infty} 2^n \lambda(E_n) = \sum_{n=1}^{\infty} 2^n (2^{-n/\alpha} - 2^{-(n+1)/\alpha}) \\ &= \sum_{n=1}^{\infty} (2^{n(1-1/\alpha)} - 2^{n(1-1/\alpha)-1/\alpha}) \\ &= \sum_{n=1}^{\infty} (1 - 2^{-1/\alpha}) (2^{n(1-1/\alpha)}) \end{aligned}$$

is geometric with ratio $2^{1-1/\alpha}$, so this series is finite if and only if $|2^{1-1/\alpha}| < 1$, which only holds when $1 - \frac{1}{\alpha} < 0$. Since α is positive we can multiply through by α and rearrange to get that $\alpha < 1$. By part (b), the finiteness of this series is equivalent to the finiteness of $\int_{(0,1)} x^{-\alpha} \, d\lambda$, so if $0 < \alpha < 1$ then the integral is finite.

The cases above show that the integral $\int_{(0,1)} x^{-\alpha} \, d\mu$ is finite if and only if $\alpha < 1$, as desired. $\square$